

STAT 24410/30030 Midterm Review Problems

1. Let X be uniform on $[0, 2]$. Conditional on $X = x$, let Y be uniform on $[0, x]$. Find the joint and marginal distributions of X and Y .

2. Suppose X is a Binomial random variable with parameters n and p , where n is known and p is not. Recall that $V[X] = np(1 - p)$. Consider the following estimator for $V[X]$:

$$\hat{V} = n \left(\frac{X}{n} \right) \left(1 - \frac{X}{n} \right).$$

(Note that there is only one sample X : n does NOT represent the number of samples, but the parameter of the binomial.)

(a) Show that the bias of $\hat{V}$ is $-p(1 - p)$.

(b) Using the result in part (a), give an unbiased estimator for $V[X]$.

3. Let X and Y be random variables with joint density given by $f(x, y) = cx, 0 \leq x, x^2 \leq y \leq 1$, and $f(x, y) = 0$ otherwise.

(a) Find the value of c for which $f(x, y)$ is a valid density.

(b) Find $E[X]$.

4. A hat store earns \$200 a day on average with a variance of \$50.

a) Give a nontrivial bound on the probability that the hat store earns at least \$250 tomorrow.

b) Give a nontrivial bound on the probability that the store will earn more than \$300 or less than \$100 (inclusive) tomorrow.

c) 5 shoppers enter the hat store. Each shopper independently decides to either buy a hat (probability $1/10$) or leave without making a purchase ($9/10$). Name the distribution that describes the number of hats sold and find the probability that the store sells at least 4 hats.

d) The hat store averages 5 visitors each day under a Poisson distribution. What is the probability that it receives exactly 2 visitors in two days?

5. You are worried that I eat too many chocolate chip cookies. On any given day, I eat 2 cookies with probability p , 1 cookie with probability q , and no cookies otherwise. (You can assume $0 \leq p, q \leq 1$ and $p + q \leq 1$, and that the days are independent.) You write down how many cookies X_i (for the i -th day) I eat over the following n days.

(a) Find the method of moments estimators for p and q .

(b) Show that your estimators are unbiased.

(c) Assume that I tell you the true value of q . Find the MLE for p .

(d) Find the estimators for p and q which jointly maximize the likelihood.

6. Unbiased estimators for the uniform distribution. Suppose that X_1 and X_2 are independent draws from $\text{Uniform}[a, b]$. It is known that the mean of this distribution is $(a + b)/2$ and the variance is $(b - a)^2/12$. If we were looking at a $\text{Uniform}[0, 1]$ distribution, recall from class that the order statistics, $X_{(1)} = \min\{X_1, X_2\}$ and $X_{(2)} = \max\{X_1, X_2\}$, have densities $f(x) = 2 - 2x$ for the min and $f(x) = 2x$ for the max (both supported on $[0, 1]$).

- (a) For $a = 0$ and $b = 1$, i.e. the $\text{Uniform}[0, 1]$ distribution, calculate $E(X_{(1)})$ and $E(X_{(2)})$.
- (b) Now let $a < b$ be any constants in the real line. Calculate the expected values $E(X_{(1)})$ and $E(X_{(2)})$. [Hint: you can think of $\text{Uniform}[a, b]$ as a linear transformation of $\text{Uniform}[0, 1]$.]
- (c) Using your work above, find some function of X_1 and X_2 such that it is an unbiased estimator of the parameter b .

7. A biased coin, which lands heads with probability $1/10$ each time it is flipped, is flipped 200 times consecutively. Let X denote the number of heads obtained in this experiment. We want to derive an upper bound on the probability X is at least 120, $P(X \geq 120)$.

- (a) Use Markov's Inequality to show that

$$P(X \geq 120) \leq \frac{1}{6}.$$

- (b) Use Chebychev's Inequality to find a different upper bound for $P(X \geq 120)$.

8. Let X have density function

$$f(x) = \frac{1}{2}(1 + x), \quad -1 < x < 1.$$

Find the density function for $Y = X^2$.

9. Sample mean and sample variance. Suppose that $X_1, \dots, X_n$ are i.i.d. draws from a distribution with mean μ and variance σ^2 . The sample variance S^2 is a function of the centered values, $X_i - \bar{X}$.

- (a) Prove that $\bar{X}$ is uncorrelated with $X_i - \bar{X}$.
- (b) We have learned before that if variables A and B are independent, then A is also independent from any function of B , i.e. A is independent of $g(B)$. However, the same does not hold for correlation/covariance: it's possible to have $\text{Corr}(A, B) = 0$ but $\text{Corr}(A, g(B)) \neq 0$. Construct a simple example to show that this can occur.
- (c) Let $X_1, X_2 \sim \text{Bernoulli}(p)$ independently, where $0 < p < 1$. Calculate $\text{cov}(\bar{X}, S^2)$. Is it zero or nonzero (your answer might depend on p)? Are $\bar{X}$ and S^2 independent or not independent (your answer might depend on p)? (It might be easiest to just make a table of all the possibilities and calculate $\bar{X}$ and S^2 for each case.)